HARVESTLINK

A Digital Platform for Pre-Harvest Forward Contracts and Micro-Yield Parametric Insurance for Smallholder Agribusiness

Document Title: Project Charter (Baseline Specification)

Document Version: 1.0

Project Phase: Requirements Engineering & MVP Scoping

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Target Domain: Agribusiness Supply Chain & InsurTech (Central, Uva, and North Central Farming Belts)

Date: October 2026

Version	Date	Author	Description of Changes	Approved By
v0.1	Initial Draft	Dilshani Rathnayake	Baseline scope and problem definition	Project Lead
v1.0	Final Charter	Dilshani Rathnayake	Added stakeholder mapping, scope boundaries, and KPIs	Academic Evaluator

1. Executive Summary & Problem Statement

1.1 The Business Problem

Smallholder vegetable and cash-crop farmers in Sri Lanka face severe systemic vulnerabilities across their production cycles:

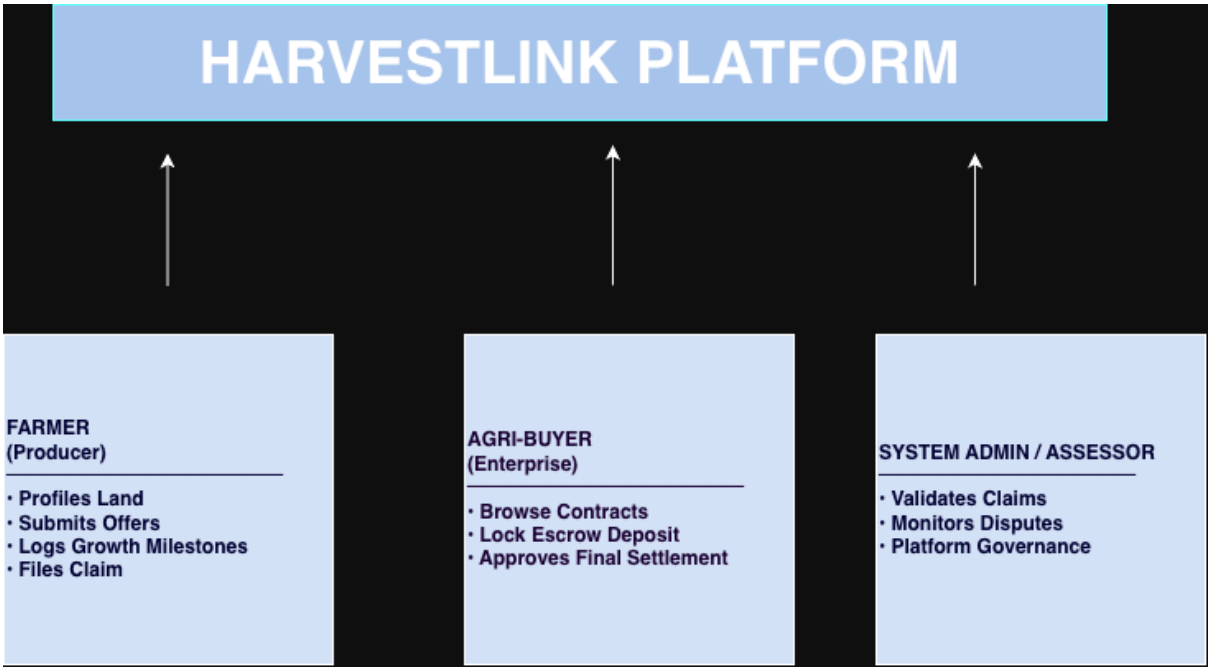
- **Severe Price Volatility & Intermediary Exploitation:** At major regional exchange hubs (such as the Dambulla Dedicated Economic Centre), farmers are subjected to unpredictable daily auction pricing, high broker commissions, and sudden market gluts, frequently compelling them to sell produce below cost.
- **Climate Risk Exposure & Delayed Relief:** Sudden unseasonal rainfall, localized flooding, and drought patterns repeatedly wipe out seasonal yields. Traditional crop insurance mechanisms require weeks or months of manual on-site physical adjustments, leaving smallholder households in crippling debt.
- **Supply Chain Inefficiencies for Institutional Buyers:** Commercial food retailers, supermarket chains (e.g., Keells, Cargills), and agro-exporters encounter erratic procurement volumes, unreliable quality grading, and complete lack of visibility into field cultivation progress.

1.2 Proposed Solution

HarvestLink bridges this divide by establishing:

1. A **digital marketplace for forward supply agreements** with guaranteed floor prices established prior to planting.
2. A **milestone-based crop validation module** that allows smallholders to submit verifiable photo updates at critical growth intervals.
3. An **automated micro-indemnity insurance engine** that evaluates localized rainfall anomalies and triggers accelerated recovery disbursements to affected farmers.

2. Stakeholder Identification & Role Mapping



Farmer (Primary User): Registers land extent, crop category, target harvest week, and minimum price threshold; captures milestone status photos using a smartphone.

Commercial Buyer (Procurement User): Discovers verified upcoming produce batches, locks supply contracts via an escrow balance guarantee, and inspects quality upon dispatch.

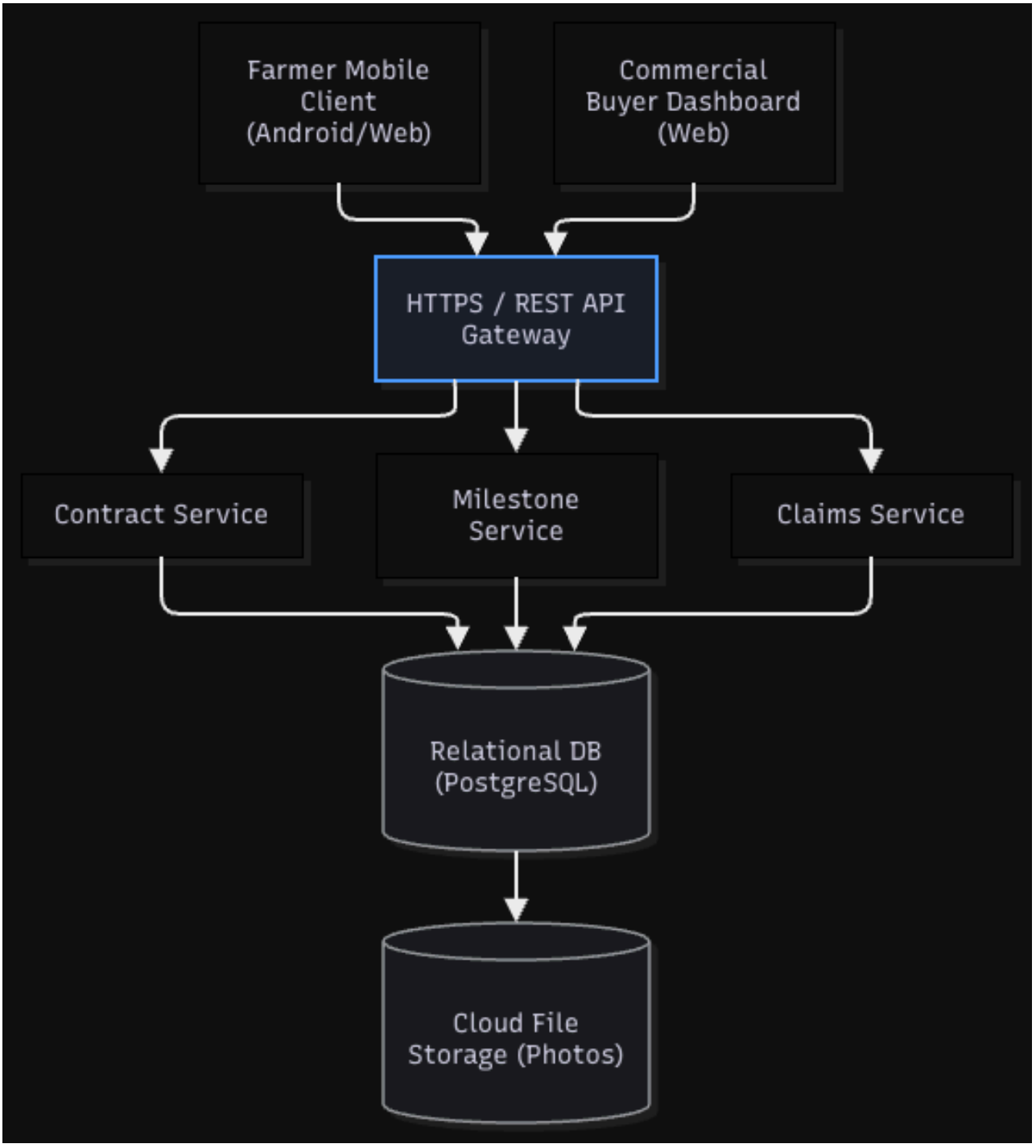
Insurance Assessor / Platform Administrator: Oversees climate anomaly flags, confirms micro-insurance claims, and moderates contractual dispute resolutions.

3. Project Scope Boundaries

To ensure rapid, successful MVP delivery within the target timeline, explicit operational boundaries are established

Scope Dimension	In-Scope (Phase 1 / MVP Deliverables)	Out-of-Scope (Deferred to Phase 2 & Future Releases)
Authentication & Access	Role-based registration and login (Farmer, Buyer, Admin) using email and mobile verification.	Biometric fingerprint verification; national digital ID API integration.
Contract Management	Forward agreement creation, price floor setting, digital acceptance workflow, and contract status tracking (*Draft, Reserved, Active, Harvested, Closed*).	Smart-contract execution over public decentralized blockchains (standard relational database transactions used instead).
Growth Tracking	Mobile photo upload for 3 cultivation milestones (Planting, Flowering/Vegetative, Pre-Harvest).	Automated drone aerial scanning, spectral indexing, or satellite computer vision analytics.
Insurance Claims	Manual farmer incident filing backed by localized precipitation threshold indicators.	Live continuous hardware IoT in-field soil moisture telemetry probes.
Financial Settlement	Escrow status logging and recorded bank transfer confirmations.	Direct multi-currency international forex exchange or crypto payment gateways.
Logistics	Collection point delivery confirmation logging.	Autonomous fleet routing, refrigerated freight dispatch, or fuel-tracking optimization.

4. High-Level System Architecture Overview



5. Implementation Roadmap & Target Milestones

Milestone ID	Target Phase	Key Deliverables	Estimated Duration
M1	Scope & Requirements	Approved Project Charter, Actor Matrix, and Scope Definition	Days 1–2
M2	Detailed Specifications	6 Core User Stories with Gherkin Acceptance Criteria & SRS	Days 2–3
M3	Process & Architecture	UML Use Case Diagram, Activity Flowchart, and ERD	Days 3–4
M4	Agile Execution Setup	Jira Cloud Scrum Project Setup, Backlog Sizing, Sprint 1 Launch	Days 4–5
M5	UX/UI Prototyping	4–6 Responsive Mobile/Web Wireframes in Figma	Days 5–6
M6	Final Portfolio Packaging	Public Documentation Workspace (Notion / GitHub) & CV Integration	Day 7

6. Project Success Criteria & KPIs

- **Functional Traceability:** 100% of defined business requirements map directly to verified User Stories in the Jira Backlog.
- **Agile Rigor:** Delivery of a simulated Sprint with fully estimated Story Points, assigned priority ranks, and an active burndown trajectory.
- **Prototyping Quality:** Interactive Figma mockup allowing an evaluator to traverse the complete contract lifecycle in under 3 minutes.
- **Assessment Readiness:** A portfolio structure ready for immediate inspection by internship hiring managers at enterprise software firms.

7. Authorization & Approval

By signing below, the project stakeholders confirm that this Project Charter serves as the agreed baseline for the HarvestLink platform requirements.

Dilshani Rathnayake

Lead Business Analyst & Project Coordinator

Date:----------

Signature:----------

Project Supervisor

Faculty of Computing, Sabaragamuwa University of Sri Lanka

Date:----------

Signature:----------